← Back **Security : Encrypting and Signing**
Graded Assignment • 30 min

Due Mar 20, 11:59 PM CET

1. Which of the following is true of security? 1 point

- ○ Perfect security is unachievable but you should always choose the most expensive option
- ◉ Perfect security is unachievable and requires a trade-off with cost
- ○ Perfect security is achievable but expensive
- ○ Perfect security is achievable and cheap

2. What is the difference between active and passive wiretapping? 1 point

- ○ In passive wiretapping only some of the network data is altered where in active wiretapping all of the network data is altered
- ○ In active wiretapping the network is snooped whereas in passive wiretapping the network is altered
- ○ Passive wiretapping and active wiretapping are different names for network snooping
- ◉ In passive wiretapping the network is snooped whereas in active wiretapping the network data is altered

3. Integrity is preserved if 1 point

- ○ The information you receive is from who you think it is and has not been modified since it was sent
- ○ The information you receive is probably from who you think it is and has not been modified since it was sent
- ○ Information you receive is from who you think it is
- ◉ The information you receive has not been corrupted since it was sent no matter who sent it

4. Which of the following factors has the smallest effect on the strength of a cryptosystem? 1 point

- ○ The encryption algorithm
- ○ The key length
- ○ The key distribution technique
- ◉ The data being transmitted

5. What is one possible advantage of public-key cryptosystems over secret-key ones? 1 point

- ◉ Public-key cryptosystems do not have the problem of secure key distribution
- ○ Public-key cryptosystems can transmit more data than secret-key ones
- ○ Public-key cryptosystems are always more secure than secret-key ones

6. What does it mean if a cryptosystem is symmetric-key in nature? 1 point

- ○ The key used for encryption is the from the key used for decryption but with a shared secret added to the end
- ○ The key used for encryption is a shortened version of the key used for decryption
- ○ The key used for encryption is the backward version of the key used for decryption
- ◉ The key used for encryption is the same as the key used for decryption

7. The following question is encrypted using a Caesar Cipher with a shift of 13. You can use www.rot13.com to decrypt the question. 1 point

Jub vf perqvgrq nf orvat bar bs gur vairagbef bs Rgurearg?

- ◉ Bob Metcalfe
- ○ Vint Cerf
- ○ Mitchell Baker
- ○ Tim Berners-Lee

8. The following question is encrypted using a Caesar Cipher with a shift of 13. You can use www.rot13.com to decrypt the question and answers. 1 point

Jung qbrf gur Gjvggre unfugnt #VUGF fgnaq sbe?

- ○ Vaqvtb, Uraan, Gnatrevar naq Fhasybjre
- ○ Vagreany Uvtu Grpuabybtl Fbyhgvba
- ○ Vagreaangvbany Uvtu Grpuabybtl Fheirl
- ◉ Vagrearg Uvfgbel, Grpuabybtl, naq Frphevgl

9. What is the SHA-1 hash of the string below as computed by http://www.dr-chuck.com/sha1.php 1 point

The Transport Layer does retransmission

- ○ 22e1791399edc7e55f7be8dbc7024
- ○ 7e55f7be8dbc7024bcb8830527722e1791399edc
- ○ 7024bcb8830521399edc7e55f7be8dbc7722e179
- ◉ 1399edc7e55f7be8dbc7024bcb8830527722e179

10. What does a cryptographic hash function do? 1 point

- ○ It converts input fixed-size bit strings into blocks of data
- ○ It takes a block of data and randomly changes characters to numbers
- ○ It computes the Hyperbolic Asymmetric Sine Harmonic (H.A.S.H.) for a sequence of audio data
- ◉ It takes a block of data and returns a fixed-size bit string called the hash value

11. What critical element does simple digest-based Message Signing as described in lecture depend on? 1 point

- ○ The message must be under 20 characters long
- ○ The secret should not be longer than the message
- ○ The geographic proximity of the transmitter and recipient of the message
- ◉ The sharing of a secret transported securely 'out of band'

12. What is the problem with secret key distribution via the internet? 1 point

- ○ The internet cannot handle the length of shared secret keys because they are longer than a single packet
- ◉ The communication of the secret key is insecure
- ○ The internet is too slow for sending keys
- ○ There is no problem

13. You are going to send the message below using shared secret of **IHTS**. Use http://www.dr-chuck.com/sha1.php to compute your message digest using the technique from lecture. What will the first six characters of the digest/signature that you send along with the message? 1 point

Be sure to drink more Ovaltine

- ○ 2b5473
- ○ e1c85e
- ◉ 8b4258
- ○ 44dbc4

14. Select the valid signed message from Annie if your shared secret is **IHTS**? Use http://www.dr-chuck.com/sha1.php to compute your message digests using the technique from lecture. Only the first 6 characters of the SHA1 message digest are shown below. 1 point

- ○ Meet me at the train station87fd2e
- ◉ Send money please7d47f3d4
- ○ Bring me cookies51be4e
- ○ I have not been kidnapped by pirates32f4d3